

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-00-066

WASTE DISCHARGE REQUIREMENTS
FOR
FOREST MEADOWS WASTEWATER TREATMENT AND RECLAMATION PLANT
CALAVERAS COUNTY WATER DISTRICT
ALSTON FINANCIAL, INC. AND EMC, INC.
CALAVERAS COUNTY

The California Regional Water Quality Control Board, Central Valley Region (hereafter Board), finds that:

1. The Calaveras County Water District (CCWD) submitted a Report of Waste Discharge (RWD), dated 18 June 1999, for the upgraded Forest Meadows wastewater treatment and reclamation plant (Plant). The Plant is located approximately two miles east of the town of Murphys. The Plant and majority of the property on which the Plant is located (Assessor's Parcel No. 34-052-03) is owned by CCWD. The remainder of the property on which the Plant is located, and the property which contains the leachfields (APN 34-052-02), is owned by EMC, Inc.
2. The Plant is located on Forest Meadows Road off Highway 4. It is situated on the north side of Angels Creek in Section 34, T4N, R14E, MDB&M with surface water drainage to Angels Creek, as shown in Attachment A, which is attached hereto and made part of this Order by reference.
3. Waste Discharge Requirements Order No. 74-326, adopted by the Board on 24 May 1974, prescribes requirements for discharge of treated domestic wastewater from the Plant to a community leachfield. Order No. 74-326 is neither adequate nor consistent with current plans and policies of the Board.
4. The former Plant design consisted of headworks facilities, two aerated ponds, a storage pond, an effluent pump station, and two leachfields designed for 30,000 gallons per day (gpd) each. The rated capacity of the Plant was 65,000 gpd. However, seepage has been observed below each of the leach fields at application rates greater than approximately 40,000 gpd.
5. According to Monitoring Reports submitted by CCWD, the current average daily discharge covering the dry months from May through October is 43,000 gpd. For the wet months of November through March, the average daily discharge is 62,400 gpd. Flows range from a low of approximately 28,000 gpd, during the dry season, to a high of approximately 193,000 gpd during the peak-wet season.
6. CCWD proposes to expand the treatment capacity of the Plant and to upgrade the treatment processes to provide reclaimed wastewater to the Forest Meadows Golf Course for irrigation. Reclaimed wastewater will be discharged from the Plant to a 108 acre-foot

impoundment (Storage Facility). The Storage Facility is located approximately 1800-feet from the Plant.

7. The Storage Facility (APN 34-052-18) and the Forest Meadow Golf Course (APN 34-075-01) are owned by Alston Financial, Inc. The Calaveras County Water District, Alston Financial Inc., and EMC Inc., are hereby jointly referred to as “Discharger”. All three entities are jointly responsible for ensuring compliance with these waste discharge requirements.

8. CCWD has entered into an April 1999 “*Effluent Storage and Disposal Agreement*” (Agreement) with Alston Financial, Inc. The purpose of the Agreement is to provide for storage and long-term utilization of reclaimed wastewater at the Forest Meadows Golf Course. The terms and conditions of the Agreement, which are specifically set forth in CCWD’s Resolution No. 98-40, state, in pertinent part, the following:

“Spray irrigation of effluent shall be accomplished in compliance with the applicable waste discharge permit(s).”

9. Plant upgrades consist of the following: The two aerated ponds have been converted to a complete mix basin and a sludge-settling storage basin, effluent from the sludge-settling basin will be filtered with two continuous backwash, deep-bed sand filters, and effluent from the sand filters will be disinfected by an ultraviolet (UV) light contact-chamber. Depending on the operational efficiency of the wastewater treatment system, the Discharger may install a clarification unit (DAF – dissolved air flotation) at a later date. Reclaimed wastewater will then be pumped to the golf course and stored for irrigation in the 108 acre-foot impoundment. The existing leach fields will be retained for emergency use to prevent spills from the Storage Facility during storm events or when Plant effluent does not meet Title 22 California Code of Regulation (CCR) standards. However, in no event shall the volume of wastewater disposed to the leachfields exceed the volume set forth in Discharge Specification B.3.
10. The reclamation Plant is designed to treat average dry weather flows up to 190,000 gpd and peak wet weather flows up to 280,000 gpd.
11. Reclamation effluent limits are based on the State Department of Health Services statewide reclamation criteria contained in Title 22, California Code of Regulations, Section 60301, et seq. (hereafter Title 22), which provide guidelines for the use of reclaimed water onto parks, playgrounds, schoolyards and other areas where the public has similar access or exposure.
12. Surrounding land uses are primarily rural residential with no industrial zoning in the project area.

13. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River and San Joaquin River Basins (hereafter Basin Plan), which contains water quality objectives for waters of the Basins. These requirements implement the Basin Plan.
14. Surface water runoff is to Angels Creek, a tributary to the New Melones Reservoir, which eventually empties to the Stanislaus River.
15. The beneficial uses of downstream waters from the Plant are municipal and domestic supply; agricultural supply; recreation; aesthetic enjoyment; groundwater recharge; fresh water replenishment; and preservation and enhancement of fish, wildlife, and other aquatic resources.
16. The beneficial uses of underlying groundwaters are municipal, industrial, and agricultural supply.
17. On 12 August 1998, CCWD certified an Initial Study/Mitigated Negative Declaration in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. The Project as approved will not have a significant effect on water quality.
18. The Board has reviewed the Initial Study/Mitigated Negative Declaration and concurs that the Project as approved will not have significant impacts on water quality.
19. The Board consulted with the State Department of Health Services and the Calaveras County Health Department and considered their recommendations regarding public health aspects for the use of reclaimed water.
20. This discharge is exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 2005, et seq., (hereafter Title 27). The exemption pursuant to Section 20090(b), is based on the following:
 - a. The Board is issuing waste discharge requirements,
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to Title 22 CCR, Division 4.5, and Chapter 11, as a hazardous waste.
21. The Board has notified the Discharger, and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.

22. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 74-326 is rescinded and the Calaveras County Water District, Alston Financial, Inc., and EMC, Inc., their agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.
4. The discharge shall not cause the degradation of any water supply.
5. Discharge of waste classified as hazardous, as defined in Sections 2521(a) of Title 23, CCR, Section 2510, et seq., (hereafter Chapter 15, or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.
6. Excessive irrigation with reclaimed water that results in excessive runoff of reclaimed water, or continued irrigation of reclaimed water during periods of precipitation, is prohibited.
7. Surfacing of wastewater in the leachfields is prohibited.

B. Discharge Specifications:

1. The average dry weather discharge flow rate shall not exceed 0.19 mgd.
2. The peak wet weather discharge flow rate shall not exceed 0.28 mgd.
3. The discharge flow to the leachfields shall not exceed the capacity of the leachfields (as determined by the report submitted per Provisions F.1). The leachfields shall have sufficient capacity to accommodate allowable wastewater flow as well as inflow and infiltration during the wet season.
4. The existing leachfield area will serve as a long-term Plant reliability feature. Wastewater disposal to the leachfields is permitted during periods of Plant repair, to prevent spillage at the Storage Facility, and when treated wastewater effluent does not

meet Title 22 CCR standards. Pursuant to Title 22 CCR, Section 60341(b), the leachfields shall be of sufficient capacity to provide for at least 20 days of emergency disposal capacity. The Plant may incorporate the use of the on-site emergency storage basin to meet the 20-day emergency disposal requirement.

5. The Plant and the Storage Facility shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
6. The Forest Meadows Storage Facility shall have sufficient capacity to contain all reclaimed wastewater flow, design seasonal precipitation, seasonal ancillary inflow, and infiltration during the wet season. Design seasonal precipitation shall be based on total annual precipitation using a return of 100 years, distributed monthly in accordance with historical rainfall patterns.
7. The freeboard in all ponds (at the Plant and Storage Facility) shall never be less than two feet as measured vertically from the water surface to the upper surface of the lowest adjacent dike or levee.
8. On or about **15 October** each year, the available Storage Facility capacity shall at least equal the volume necessary to comply with Discharge Specification Nos. 6 and 7.
9. Objectionable odors originating at the Plant or Storage Facility shall not be perceivable beyond the boundaries of the Plant or Storage Facility.
10. As a means of discerning compliance with Discharge Specification No. 9, the dissolved oxygen content shall not be less than 1.0 mg/l in the Storage Facility, as measured at a point as far as practical from the inlet and within one foot of the water surface.
11. Public contact with reclaimed wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.

C. Effluent Limitations:

1. The discharge to the Storage Facility of an effluent in excess of the following limits is prohibited:

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>	<u>Weekly Median</u>	<u>Daily Average</u>
Total Coliform Organisms	MPN/100 ml	-	23	2.2	-

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>	<u>Weekly Median</u>	<u>Daily Average</u>
Settleable Solids	ml/l	0.2	0.5	-	-
BOD ¹	mg/l	20	30	-	-
Turbidity	NTU	-	-	-	2 ²

¹ 5-Day, 20° Celsius Biochemical Oxygen Demand

² Not to exceed 5 NTU more that 5% of the time during a 24-hour period.

D. Reclamation Requirements:

1. Reclaimed wastewater shall meet the criteria contained in Title 22, Division 4, CCR (Section 60301, et seq.).
2. Reclaimed wastewater shall be discharged to the Forest Meadows Golf Course in accordance with a Wastewater Disposal Operations Plan to be submitted to the Executive Officer for approval.
3. Reclaimed wastewater conveyance lines shall be clearly marked as such.
4. Reclaimed water controllers, valves, etc., shall be affixed with reclaimed water warning signs, and these and quick couplers and sprinkler heads shall be of a type, or secured in a manner, that permits operation by authorized personnel only.
5. Areas irrigated with reclaimed water shall be managed to prevent breeding of mosquitoes. More specifically,
 - a. All applied irrigation water must infiltrate completely within a 12-hour period.
 - b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.
 - c. Low-pressure and un-pressurized pipelines and ditches accessible to mosquitoes shall not be used to store reclaimed water.
6. Reclaimed water for irrigation shall be managed to minimize erosion, and runoff from the disposal area.

7. Direct or windblown spray shall be confined to the designated reclamation area and prevented from contacting drinking water facilities.
8. The Discharger may not irrigate with effluent during periods of precipitation and for at least 24 hours after cessation of precipitation, or spray irrigate when wind velocities exceed 30 mph.
9. Signs with proper wording of sufficient size shall be placed at areas of access and around the perimeter of all areas used for effluent disposal to alert the public of the use of reclaimed water.
10. Runoff from the irrigation field shall not be discharged to any surface water drainage course within 24 hours of the last application of reclaimed water.
11. There shall be no impoundment of reclaimed water within 50 feet of any domestic water well or irrigation well unless it is demonstrated to the satisfaction of the Executive Officer that a shorter distance is justified.

E. Solids Disposal Requirements:

1. Collected screenings, sludge, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer, and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq.
2. Any proposed change in sludge use or disposal practice from a previously approved practice shall be reported to the Executive Officer and U.S. Environmental Protection Agency (EPA) Regional Administrator at least 90 days in advance of the change.
3. Use and disposal of sewage sludge shall comply with existing Federal, State, and local laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.
4. If the State Water Resources Control Board and the Regional Water Resources Control Board are given the authority to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. CCWD shall comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

F. Groundwater Limitations:

1. The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentration statistically greater than background water quality, except for Coliform bacteria. For coliform bacteria, increases shall not cause the most probable number of total coliform organisms to exceed 2.2 MPN/100 ml over any 7-day period.

G. Provisions:

1. By **25 April 2000**, the Discharger shall submit a report prepared and signed by a registered engineer which evaluates the available disposal capacity of the existing leachfield system, determines whether the capacity of the leachfield will need to be increased to be in compliance with Prohibition A. 7, and Discharge Specification B. 4, and provides design plans, if necessary, to increase the capacity of the existing leachfield. If the existing leachfields will not meet the requirements of Prohibition A. 7, Discharge Specification B. 4, then the Discharger must provide design plans for the construction of new leachfields.
2. By **25 July 2000**, the Discharger shall provide a report prepared and signed by a registered engineer that certifies the Plant has increased the available leachfield disposal capacity, if necessary as determined by Provision F. 1.
3. By **25 October 2000**, the Discharger shall provide a report prepared and signed by a registered engineer that certifies the Plant has increased the available leachfield disposal capacity by the construction of new leachfields, if necessary as determined by Provision F. 1.
4. By **1 May 2000**, the Discharger shall submit a Wastewater Disposal Operations Plan that describes in detail how, when, and where wastewater will be applied to the golf course.
5. By **1 May 2000**, the Discharger shall submit written verification of compliance with Provision G. 13, including a copy of each operator's certification.
6. By **1 September 2000**, the Discharger shall submit a comprehensive water balance analysis to determine compliance with Discharge Specifications B.4 and B.5. Total annual precipitation shall be based on a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. If insufficient volume is available, then the report shall also contain a plan and time schedule for coming into full compliance with this Order.

7. By **1 October 2000**, the Discharger shall submit a Solids Management Plan for the permanent disposal of biosolids, the long-term management of biosolids, and for all other non-effluent wastes generated by the treatment process. The Solids Management Plan shall provide a detailed program and schedule for permanent disposal of all solid wastes that will be generated in the future. Information provided shall include methods and locations of temporary on-site storage (if used), Best Management Practices for on-site handling and storage of solid waste, means of disposal, frequency of disposal, and disposal site (as applicable).
8. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving the disposal or reclamation areas, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
9. The Discharger shall comply with Monitoring and Reporting Program No. 5-00-066, which is part of this Order, and any revisions thereto, as ordered by the Executive Officer.
10. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which is attached hereto and made part of this Order by reference. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)".
11. The Discharger shall submit to the Board on or before each compliance report due date the specified document, or if appropriate, a written report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is reported, then the Discharger shall state the reasons for noncompliance and shall provide a schedule to come into compliance.
12. The Discharger shall use the best practicable cost-effective control technique(s) currently available to comply with discharge limits specified in this order.
13. The Discharger shall provide certified wastewater treatment plant operators in accordance with Title 23 of the California Code of Regulations, Division 3, Chapter 26.
14. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.
15. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by CCWD, Alston Financial, Inc., or EMC, Inc., then the party shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this office.

16. CCWD, Alston Financial, Inc., and EMC, Inc., shall comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
17. A copy of this Order shall be kept at the discharge facility for operating personnel. Key operating personnel shall be familiar with its contents.
18. The Board will review this Order periodically and may revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 17 March 2000.

GARY M. CARLTON, Executive Officer

Attachments

DLM: 3/17/00